

Design and Analysis of Algorithms

Course Metadata

Credit Structure	L-T-P: 4-0-0 Total: 4
Offering Semester	3 rd Semester, Autumn
Level of Subject	UG Level
Compulsory or elective?	Core
Pre-requisite	Programming & Data Structures
Faculty Names	Adway Mitra, Koustav Rudra, Plaban Kumar Bhowmick, Jiaul H Paik
Overlapping Subjects	ALGORITHMS – I (CS21003), ALGORITHMS II (CS31005), ALGORITHMS (EC31205), DESIGN & ANALYSIS OF ALGORITHMS (MA30207) ~ 100%
Justification of overlap	This course will be offered as a CORE to the BTech in AI program

Course Description

This course provides an abstract or algorithmic treatment of the computational problems. This course introduces several algorithm design paradigms and their analysis through mathematical and formal framework of computational complexity. This course also discusses several advanced data structures and their relation to different computational problems. Finally, formal treatment of problems pertaining to graph, arithmetic and geometric domains are also discussed.

Course Content

Introduction to algorithms, Algorithm Analysis Framework, Algorithm Design Paradigms - Introduction, Divide and Conquer, Greedy Strategy, Dynamic Programming, Basic Data Structures, Sorting and Order Statistics, Search Data Structures, Graph Algorithms, Geometric Algorithms, Arithmetic Algorithms, NP-Completeness

Lecture-wise Breakup

Sl No.	Module	Unit	Module Hr (48)
1	Introduction	Role of algorithms in computing, Algorithm as technology, Models of computation	2
2	Algorithm Analysis Framework	Definition and elements of algorithm, Characterizing running time, Asymptotic notations	4

3	Algorithm Design - Paradigms Introduction	Recursion, divide-and-conquer, greedy, dynamic programming.	2
4	Divide and Conquer	Elements of divide-conquer strategy, Matrix Multiplication, Merge sort, Closest Pair	4
5	Greedy Strategy	Elements of greedy strategy, Huffman coding, Travelling salesman problem, Fractional knapsack problem	4
6	Dynamic Programming	Optimization problems, Matrix chain multiplication, Elements of dynamic programming, Longest common subsequence, Minimum Edit distance, Knapsack problem	4
7	Basic Data Structures	Stacks and queues, graphs and trees, binary trees	2
8	Sorting and Order Statistics	Heap sort, Quicksort, Sorting in linear time, Medians and Order Statistics	4
9	Search Data Structures	Binary search trees, height balancing, B-trees, skip lists, hashing.	6
10	String Matching	Rabin-Karp algorithm, Knuth-Morris-Pratt Algorithm, Boyer-Moore Algorithm, Shift-AND Matching Algorithm	4
11	Graph Algorithms	Breadth-First search, Depth-first search, topological ordering, Minimal Spanning trees, Single-source shortest paths, All-Pairs Shortest Paths	8
12	NP-Completeness	Classes P and NP, reduction, NP-completeness, examples of NP-complete problems	4

Books

1. **Introduction to Algorithms**, Thomas H Cormen, Charles E Lieserson, Ronald L Rivest and Clifford Stein, 4th Edition, MIT Press, 2022
2. **Algorithm Design**, Jon Kleinberg and Éva Tardos, 1st Edition, Pearson, 2005
3. **Algorithms -- A Creative Approach**, Udi Manber Pearson; 1st edition. 1989

References

1. **Algorithms**, Sanjoy Dasgupta, Christos H. Papadimitriou and Umesh V. Vazirani, Tata McGraw-Hill, 2008.
2. **Algorithm Design: Foundations, Analysis, and Internet Examples**, Michael T Goodrich and Roberto Tamassia, Second Edition, Wiley, 2006.

3. **Computational Geometry: Algorithms and Applications**, Mark de Berg, Mark van Kreveld, Mark Overmars and Otfried Shwarzkopf (Cheong), Third edition, Springer-Verlag, 2008.
4. **Algorithms: Design Techniques and Analysis**, Muhammad H. Alsuwaiyel, World Scientific, 1999.
5. **Computer Algorithms: Introduction to Design and Analysis**, Sara Baase and Allen Van Gelder, third edition, Addison-Wesley, 2000.